

Node.js Developer (4 Years Experience) – Complete Skill Checklist

1. JavaScript Fundamentals

- Closures
 - Scope
 - Hoisting
 - Prototypes
 - Event Loop
 - Call Stack
 - Microtask Queue
 - Callback Queue
 - Promises
 - Async/Await
 - ES6+ Features
 - Array Methods (map, filter, reduce)
 - Destructuring
 - Spread Operator
 - Rest Parameters
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2. Node.js Core

Modules

- require()
- module.exports
- CommonJS
- ES Modules

File System

- fs.readFile()
- fs.writeFile()
- fs.createReadStream()
- fs.createWriteStream()

Events

- EventEmitter
- Custom Events

Streams

- Readable Streams
- Writable Streams
- Transform Streams
- Pipe

Buffers

- Buffer.from()
 - Buffer.alloc()
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3. Express.js

- Routing
 - Route Parameters
 - Query Parameters
 - Middleware
 - Error Handling Middleware
 - Request Lifecycle
 - Validation
 - Logging
 - API Versioning
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4. Authentication & Security

JWT

- jwt.sign()
- jwt.verify()

Password Hashing

- bcrypt

Security Concepts

- Helmet
- CORS
- XSS Protection
- CSRF Protection
- Rate Limiting

Authentication vs Authorization

5. Database

MongoDB

- CRUD Operations
- Aggregation Pipeline
- Lookup
- Indexing
- Transactions

SQL (MySQL/PostgreSQL)

- SELECT
 - INSERT
 - UPDATE
 - DELETE
 - INNER JOIN
 - LEFT JOIN
 - GROUP BY
 - HAVING
 - Subqueries
 - Indexes
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6. REST API Design

HTTP Methods

- GET
- POST
- PUT
- PATCH
- DELETE

Status Codes

- 200 OK
- 201 Created
- 400 Bad Request
- 401 Unauthorized
- 403 Forbidden
- 404 Not Found
- 500 Internal Server Error

Features

- Pagination

- Search
 - Sorting
 - Filtering
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7. Error Handling

- try/catch
 - Custom Error Classes
 - Global Error Middleware
 - Async Error Handling
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8. Performance Optimization

Caching

- Redis

Compression

- compression()

Database Optimization

- Indexing
- Query Optimization

Memory Management

- Detect Memory Leaks
 - Optimize Resource Usage
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9. Project Architecture

MVC Pattern

- Model
- View
- Controller

Service Layer Pattern

Repository Pattern

Folder Structure

controllers/ services/ routes/ middlewares/ models/ utils/ config/

10. Testing

Unit Testing

- Jest

API Testing

- Supertest

Mocking

Test Coverage

11. DevOps Basics

- Git
- GitHub
- Branching Strategy
- CI/CD
- Docker Basics
- Environment Variables

Example:

PORT=3000 JWT_SECRET=secret

12. Real-World Project Features

- User Authentication
- Role-Based Access Control (RBAC)
- CRUD APIs
- File Upload
- Email Service
- Payment Gateway Integration

- Third-Party API Integration
 - Background Jobs
 - Notification System
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13. System Design Basics

- Caching
 - Load Balancers
 - Database Scaling
 - Message Queues
 - API Gateway
 - URL Shortener Design
 - High Traffic API Design
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Interview Readiness Checklist

- ☐ JavaScript Advanced Concepts
- ☐ Node.js Core Modules
- ☐ Express.js Advanced APIs
- ☐ JWT Authentication
- ☐ MongoDB / SQL Databases
- ☐ REST API Design
- ☐ Error Handling
- ☐ Redis Caching
- ☐ Testing (Jest)
- ☐ Docker Basics
- ☐ Git & CI/CD
- ☐ System Design Basics

Completion of all topics = Strong 4-Year Node.js Developer Interview Readiness.